HW 6

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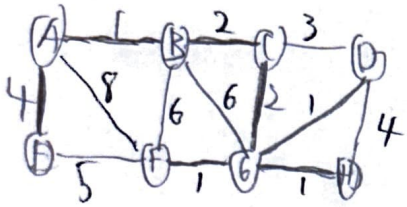
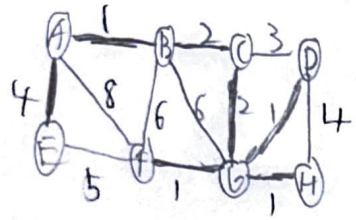
1.



7 edges
8-1

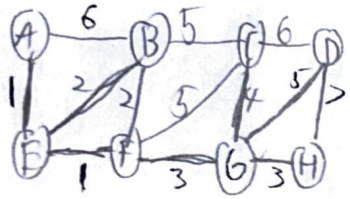
kruskal: AB, DG, FG, GH, BC, CG, AE



7 edges
8-1 ✓

Prim: AB, BC, CG, DG, FG, GH, AE

2.



MST potential diffs

EB vs FB inclusion

You can have 2 potential MSTs depending on if you choose edge FB or EB. Cost is 1+1+2+3+3+4+5 = 19.

3.

best case $m=n$

worst case $m=n^2$

m = edges

n = vertices

kruskal DFT for cycle det. $\Theta(mn)$

kruskal disjoint set for cycle det. $\Theta(m\log(m))$

Prim LL $\Theta(n^2)$

Prim Binary Heap $\Theta((m+n)\log(n))$

kruskal DFT worst case $\Theta(nm), m=n^2$ $\approx \Theta(n^3)$ >> kruskal DFT Best case $\Theta(nm), m=n$ $\approx \Theta(n^2)$ == Prim LL $\Theta(n^2)$

depending on density of graph Prim LL can be faster than Prim BH and kruskal Disjoint sets

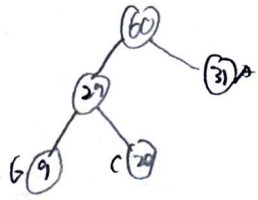
Prim Binary Heap $\Theta((m+n)\log(n))$ $\approx$ $\Theta(m\log n)$ == kruskal DisJ. Set $\Theta(m\log(m))$ $\approx$ (worst case $m=n^2$) $\Theta(m\log(n^2)) = \Theta(m2\log(n)) \approx \Theta(m\log(n))$

4.

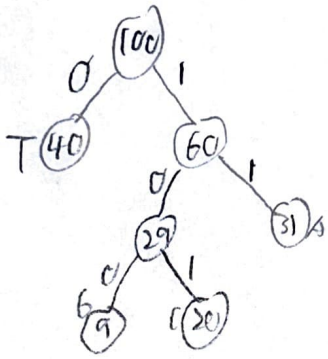
	freq
A	31
C	20
G	9
T	40

20+9=29

29+31=60



40+60=100



T: 0
A: 11
G: 100
C: 101

31% (3 bil) = .93 bil : A

9% (3 bil) = .27 bil : G

20% (3 bil) = .6 bil : C

40% (3 bil) = 1.2 bil : T

1.2 bil (1) + .93 bil (2) + (3)(.27 + .6) =

5.67 bil bits or

.70875 GB